

# Deterministic geometric Savitzky-Golay dynamics estimator

Shared parameter update: reference -> estimated

|                       |                                     |                                   |
|-----------------------|-------------------------------------|-----------------------------------|
| mass [kg]             | 2.351597591 -> 2.366278399          | delta=+0.0146808                  |
| ratio=1.0062429       |                                     |                                   |
| CoG x [m]             | -0.002024708562 -> -0.00266380413   | delta=-0.000639096                |
| ratio=-               |                                     |                                   |
| CoG y [m]             | -3.052657894e-05 -> 0.0008609471424 | delta=+0.000891474                |
| ratio=-               |                                     |                                   |
| CoG z [m]             | 0.009509749599 -> 0.0104679079      | delta=+0.000958158                |
| ratio=-               |                                     |                                   |
| rotor effectiveness 1 | 1 -> 0.9623718019                   | delta=-0.0376282 ratio=0.9623718  |
| rotor effectiveness 2 | 1 -> 0.9734229066                   | delta=-0.0265771 ratio=0.97342291 |
| rotor effectiveness 3 | 1 -> 0.9593407418                   | delta=-0.0406593 ratio=0.95934074 |
| rotor effectiveness 4 | 1 -> 0.9518064296                   | delta=-0.0481936 ratio=0.95180643 |
| rotor command lag [s] | 0.004997730255 -> 0.190958181       | delta=+0.18596                    |
| ratio=-               |                                     |                                   |

Inertia matrix [kg m^2]

reference:

```
[ 0.06500006148, -7.27899253e-07, 1.901508003e-05]
[-7.27899253e-07, 0.06495265634, 5.9167305e-08]
[ 1.901508003e-05, 5.9167305e-08, 0.1289921107]
```

estimated:

```
[ 0.06501164592, -9.881860614e-06, 2.314021975e-05]
[-9.881860614e-06, 0.06524281698, 6.923741363e-06]
[ 2.314021975e-05, 6.923741363e-06, 0.1289946473]
```

Principal moments: reference -> estimated

|                             |                                |                    |
|-----------------------------|--------------------------------|--------------------|
| principal moment 1 [kg m^2] | 0.06495264516 -> 0.06501121571 | delta=+5.85705e-05 |
| ratio=1.0009017             |                                |                    |
| principal moment 2 [kg m^2] | 0.06500006701 -> 0.06524323808 | delta=+0.000243171 |
| ratio=1.0037411             |                                |                    |
| principal moment 3 [kg m^2] | 0.1289921163 -> 0.1289946564   | delta=+2.54012e-06 |
| ratio=1.0000197             |                                |                    |

Optimization

joint dynamics loss: 1.14287225726  
Gaussian-prior cost: 0.360301973025  
selected strict-ZOH rotor lag [s]: 0.190958180968

Bag failure\_1\_single

reconstruction position RMSE [m]: 0.00156560090701  
reconstruction position component RMSE [m]: [0.000908 0.000474 0.001184]  
reconstruction orientation RMSE [deg]: 0.845162170593  
gyro consistency RMSE xyz [rad/s]: [0.051211 0.102385 0.030996]  
specific-force consistency RMSE xyz [m/s^2]: [0.944728 0.146922 0.281552]  
inferred external force RMS xyz [N]: [0.831824 0.391262 3.046729]  
inferred external torque RMS xyz [N m]: [0.075922 0.245766 0.097857]  
bag dynamics loss: 1.14287225726  
gimbal command lag [s] 0.004998683929 -> 0.03785659922

Recorded command timestamp intervals (data-derived)

Bag failure\_1\_single

rotor: median dt=0.00499773026 s (200.091 Hz), min=2.00271606e-05 s, max=0.0138893127 s

gimbal: median dt=0.00499868393 s (200.053 Hz), min=4.529953e-06 s, max=2.48282003 s

Selected optimizer diagnostics

status=1 success=True nfev=4 njev=4  
message='gtol' termination condition is satisfied.  
reported optimality=2.7666576585261282e-08  
initial cost=1.5031743504563846  
final cost=1.5031742302822693  
cost reduction=1.2017411532561084e-07  
coordinate step L2=5.11637436405372e-05  
final gradient inf-norm=2.7666576585261282e-08  
gradient tolerance satisfied=True  
final Jacobian rank=14  
final Jacobian cond(nonzero)=410.1436198608981  
max directional finite-difference Jacobian relative error=9.027424324664627e-12